

Data Sources — Shadow Node Theory

Case 1: Bruges → Antwerp (1300-1589)

Variable: Total estimated population (thousands of inhabitants)

Uncertainty: $\pm 20\%$ for medieval estimates

Year	Bruges	Antwerp	Source
1300	46,000	5,000	Nicholas (1992)
1375	42,000	10,000	Nicholas (1992)
1450	38,000	18,000	Nicholas (1992)
1500	30,000	45,000	Van der Wee (1963)
1520	22,000	65,000	Van der Wee (1963)
1560	15,000	105,000	Gelderblom (2013)
1589	8,000	42,000*	Gelderblom (2013)

*Antwerp 1589: post-Spanish Fury collapse. Excluded from power law fit (exogenous shock).

Primary references:

- Nicholas, D. (1992). *Medieval Flanders*. Longman, London.
- Van der Wee, H. (1963). *The Growth of the Antwerp Market and the European Economy*. Martinus Nijhoff, The Hague.
- Gelderblom, O. (2013). *Cities of Commerce: The Institutional Foundations of International Trade in the Low Countries, 1250-1650*. Princeton University Press.

Case 2: Toledo → Madrid (1528-1787)

Variable: Total population (inhabitants)

Uncertainty: $\pm 15\%$ for 1600-1750 Ringrose interpolations

Year	Toledo	Madrid	Source
1528	31,930	5,000	Castilian Census 1528
1561	56,270	15,000	Castilian Census 1561
1600	40,000*	55,000	Ringrose (1973) est.
1661	25,000	120,000	Ringrose (1973)
1787	15,000*	156,672	First Official Spanish Census 1787

*Interpolated estimates based on parish records analyzed by Ringrose.

Primary references:

- Ringrose, D.R. (1973). Madrid and the Spanish Economy, 1560-1850. *Journal of Economic History*, 33(2), 284-314.
- Instituto Nacional de Estadística, España. Historical population series.
<https://www.ine.es>

Case 3: Portugal vs NW Europe (1535-1980)

Variable: GDP per capita in international dollars, 2011 PPP

Source: Maddison Project Database 2023

Year	Portugal	Netherlands	UK	NW avg	Ratio NW/PT
1535	1,290	3,111	1,665	2,388	1.851
1600	1,258	4,270	1,691	2,981	2.368
1700	1,572	3,377	2,412	2,895	1.841
1750	2,184	3,777	2,702	3,240	1.483
1800	1,459	4,184	3,343	3,764	2.580
1913	1,992	6,454	8,212	7,333	3.681
1980	12,822	23,438	20,612	22,025	1.718

Note on NW Europe definition: Average of Netherlands and United Kingdom,

consistent with Costa, Palma & Reis (2015) who identify this core as the benchmark for Portugal's convergence analysis.

Primary references:

- Bolt, J. and van Zanden, J.L. (2024). Maddison Project Database 2023. University of Groningen. <https://www.rug.nl/ggdc/historicaldevelopment/maddison/>
- Costa, L.F., Palma, N., and Reis, J. (2015). The great escape? The contribution of the empire to Portugal's economic growth. *European Review of Economic History*, 19(1), 1-22. https://cooperative-individualism.org/costa-leonor_portugal's-economic-growth-2015-feb.pdf
- Allen, R.C. (2001). The Great Divergence in European Wages and Prices. *Explorations in Economic History*, 38(4), 411-447.

Case 4: Tlaxcala → Puebla (1550-2022)

Variable: GDP per capita relative index (Tlaxcala = 100 in 1535)

Construction method:

Pre-1993 (estimated):

- Maddison Project Mexico national GDP per capita series used as proxy for regional differential
- Two calibration anchors:
 - 1535: estimated 12% initial Puebla advantage (post-Real Cédula designation as principal city)
 - 1993: verified 48.8% advantage from INEGI direct data (ratio = 1.49)
- Amplification periods: bypass railway construction (1873), Puebla industrialization (1940-1980)

Post-1993 (direct INEGI data):

Year	Puebla/Tlaxcala ratio	Source
1993	1.490 (+48.8%)	INEGI SCNM
2000	1.520	INEGI SCNM
2010	1.540	INEGI SCNM

2018	1.570	INEGI SCNM
2022	1.550	INEGI SCNM

Additional verified metrics (INEGI 2022):

- Migration-divergence correlation: $r = 0.9646$
- FDI ratio Puebla/Tlaxcala: 25.5×
- Labor outflow: 14.5% of Tlaxcalan workforce employed outside the state

Primary references:

- INEGI (2022). PIB per cápita por entidad federativa. Sistema de Cuentas Nacionales de México. <https://www.inegi.org.mx/temas/pib/>
- CONEVAL (2022). Indicadores de pobreza y bienestar por entidad federativa.
- Secretaría de Economía (2022). Inversión Extranjera Directa por entidad federativa.
- Maddison Project Database 2023 (Mexico series, 1550-1820).

Case 5: HackerEarth Elite vs Basic (2026)

Dataset: zerve_hackathon_dataset.csv

Status: Proprietary — not redistributable

Size: ~299 MB, ~4.77M rows, 4,774 unique users, 141 event types

Aggregate results available in: shadow_node_maddison_resumen.csv

Metric	Elite (0.5%)	Basic (93.1%)	Ratio
VDR mean	47.86	0.0063	7,478×
Active days	30.9	1.2	25×
Tool diversity	8.5	0.08	106×
Credits used	high	near zero	~6,000×
CSI V3 range	55.62–74.76	0–18.57	—

Model metrics:

- Churn classifier (GradientBoostingClassifier, 284 features): ROC-AUC = 0.9994
- Cross-validation (5-fold stratified): ROC-AUC = 1.0000
- Top retention predictor (SHAP): agent_accept_suggestion (~0.50 importance)

Reference:

- Captain 1n2a1n05 (2026). Fractal Core Framework V3. HackerEarth Hackathon 2026.